Modeling Bimanual Perceptual Interactions in Macaque Monkeys

Bayesian Hierarchical Modeling

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Outline

Task and Data

Model A: Sessions Pooled

Model B: Sessions Hierarchical

Model C: Distractor Amplitude as a Covariate

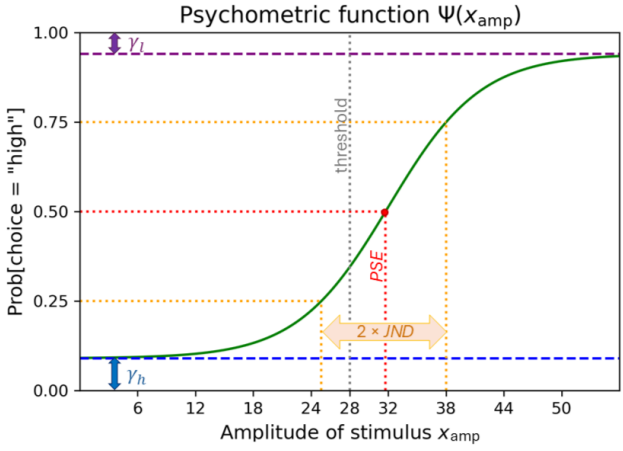
Summary

The task: 2AFC categorization of vibration amplitude

- stimulus $x_{\text{stim}} \in \{6, 12, 18, 24, 32, 38, 44, 50\}\mu m$
  - administered to one finger of one hand (left or right)
- distractor $x_{\text{dist}} \in \{0, x_{\text{d}}\}\mu m$
  - $x_{\text{d}} \in \{6, 12, 18, 24, 32\}\mu m$ value fixed for each session
  - administered to the other (non-stimulus) hand
- if present, distractor administered simultaneously with the stimulus
- monkey categorizes the stimulus as either "high" or "low" relative to a trained boundary of 28 μm.
  - "low" response: correct for $x_{\text{stim}} < 28\mu m$
  - "high" response: correct for $x_{\text{stim}} > 28\mu m$.
- correct responses are rewarded with water or juice

Psychometric function: response $r \sim \text{Bernoulli}(p = \psi)$

$$\psi(x_{\text{stim}}) = \gamma_h + (1 - \gamma_h - \gamma_l)\ \text{logit}^{-1}(\beta_0 + \beta_1 x_{\text{stim}})$$

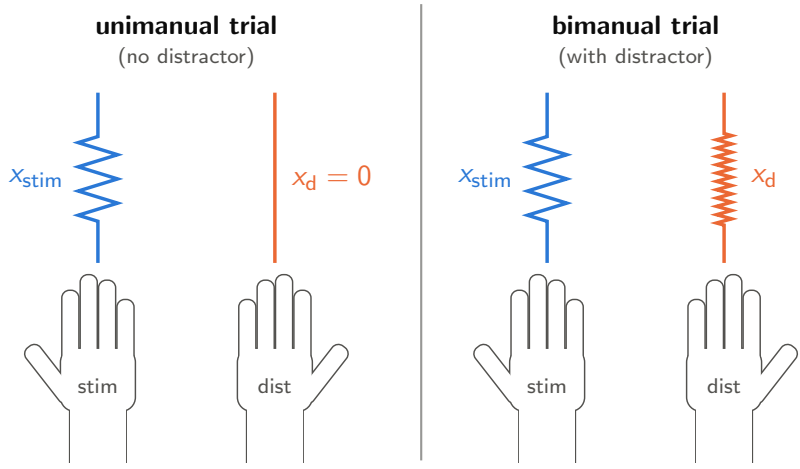


Parameters

$$\psi(x_{\text{stim}}) = \gamma_h + (1 - \gamma_h - \gamma_l)\ \text{logit}^{-1}(\beta_0 + \beta_1 x_{\text{stim}}), \qquad \vec{\theta} = (\beta_0, \beta_1, \gamma_h, \gamma_l)$$

- γ_h: lapse rate for "high"
  - lower bound on $\mathbf{P}(\text{guess} = \text{"high"})$, even to the weakest stimuli
  - lower (left) asymptote of ψ
- γ_l: lapse rate for "low"
  - lower bound on $\mathbf{P}(\text{guess} = \text{"low"})$, even to the strongest stimuli
  - $1 - \gamma_l$ = upper (right) asymptote of ψ
- logit^{-1}: the logistic sigmoid, $\text{logit}^{-1}(z) = \dfrac{1}{1 + \exp(-z)}$
- PSE: stimulus where responses of "high" and "low" are equally likely
  - x_{stim} such that $\psi(x_{\text{stim}}) = 0.5$
  - represents bias
- JND: how much the stimulus needs to change to be noticed half the time
  - half the distance between $\psi(x_{\text{stim}}) = 0.25$ and 0.75
  - represents sensitivity

The task: one trial



Modeling bimanual interactions

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  - $x_{\text{d}} \in \{6, 12, 18, 24, 32\}\mu m$ value fixed for each session
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The data

- One monkey (Sirius), 42 sessions, 12,937 trials.
- 8 stimulus amplitudes, 6–50 μm; 4 groups: hand $\times$ (unimanual, bimanual).
- Each session uses a single distractor amplitude (6–32 μm) on its bimanual trials.

		left hand		right hand		
x_d (μm)	sessions	uni	bi	uni	bi	trials
6	8	572	576	575	575	2,298
12	6	482	480	479	477	1,918
16	2	158	154	157	159	628
18	1	80	80	80	80	320
20	1	80	78	80	80	318
22	1	78	78	78	79	313
24	7	557	556	554	558	2,225
26	1	79	80	80	80	319
28	1	80	78	80	80	318
30	1	79	73	76	72	300
32	13	992	997	997	994	3,980
total	42	3,237	3,230	3,236	3,234	12,937

Three models

Model	Sessions	Distractor
A	pooled: one curve per group	uni and bi fit separately
B	hierarchical: one curve per group and session	uni and bi fit separately
C	hierarchical, one group per hand	covariate: shift $\beta_2 d$

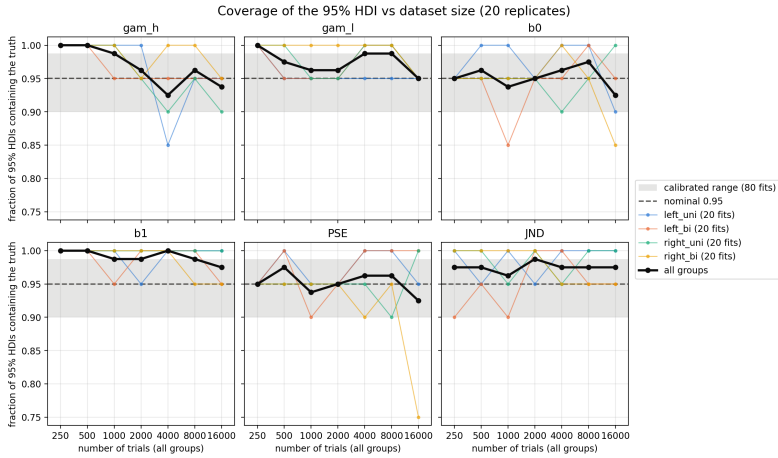
For each model: convergence study on synthetic data, then the fit to the real data.

Model A

- One psychometric curve per group (hand $\times$ condition), shared by all 42 sessions: 16 parameters.
- Priors: $\beta_0 \sim \mathcal{N}(0, 6)$, $\beta_1 \sim \mathcal{N}(4, 2)$, $\gamma = 0.25\, w$ with $w \sim$ Beta.

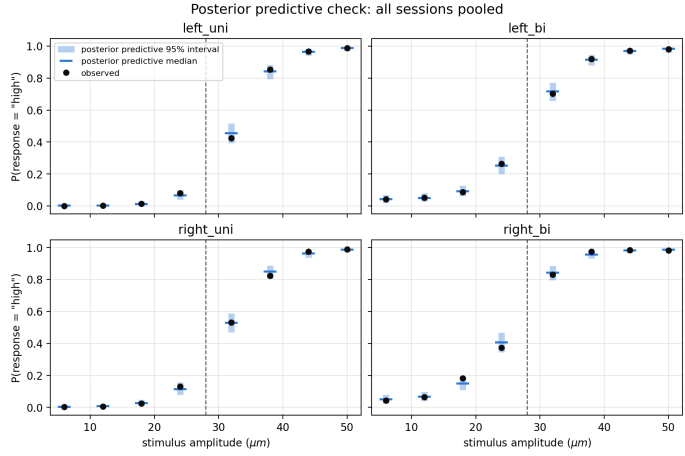
TODO: one-line motivation: simplest model, a baseline

Model A: convergence on synthetic data



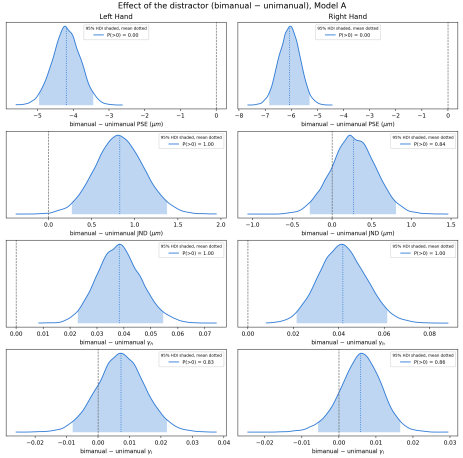
TODO: takeaway; numbers from Convergence_A_Report, "Conclusions"

Model A: fit to the real data



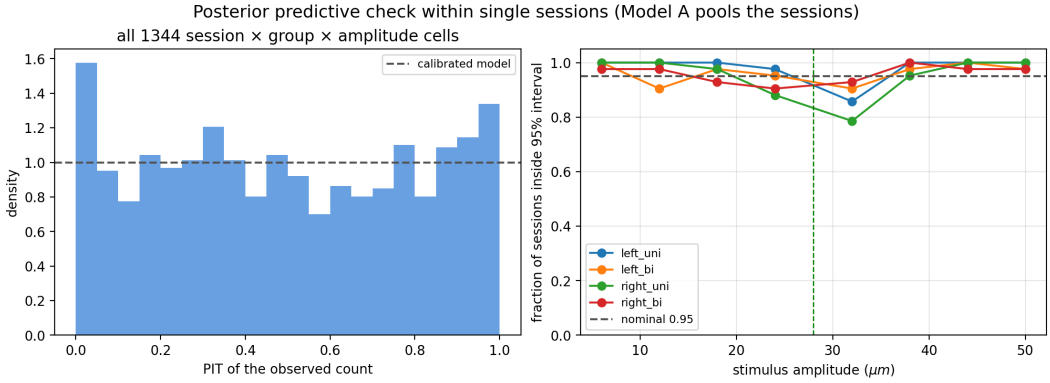
TODO: pooled check passes: all 32 proportions inside (Fit_A_Report, "Model Checks")

Model A: effect of the distractor



TODO: PSE shift, γ_h and JND effects (Fit_A_Report, Summary)

Model A misses the variation between sessions



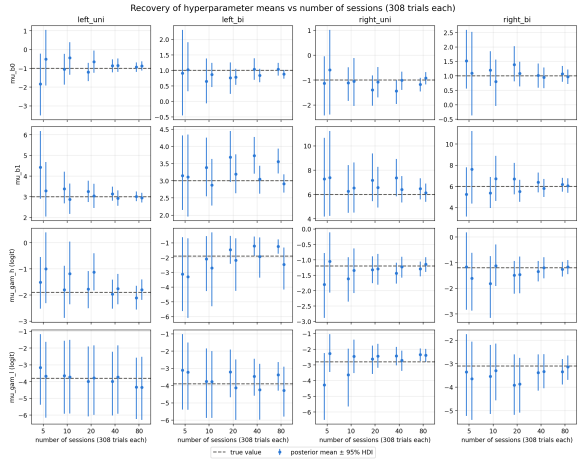
TODO: within single sessions responses vary more than A predicts (Fit_A_Report, "Model Checks")

Model B

- A curve for every group *and session*, drawn from group-level distributions (means and between-session sds).
- Design-based priors, checked by prior predictive simulation.

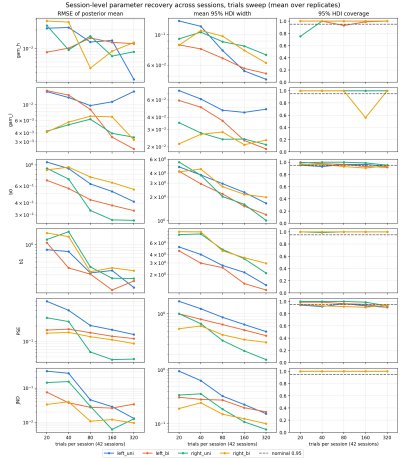
TODO: plate diagram or equations (Convergence_B_Report, "Model Structure")

Model B: convergence on synthetic data



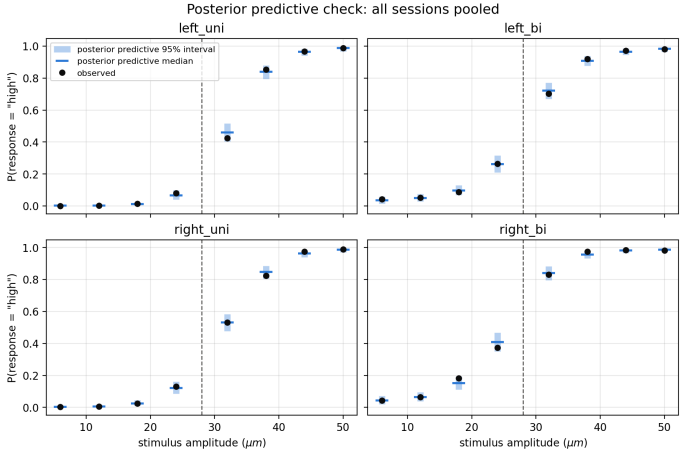
TODO: what is recovered at the real data size (Convergence_B_Report, "Conclusions")

Model B: per-session recovery



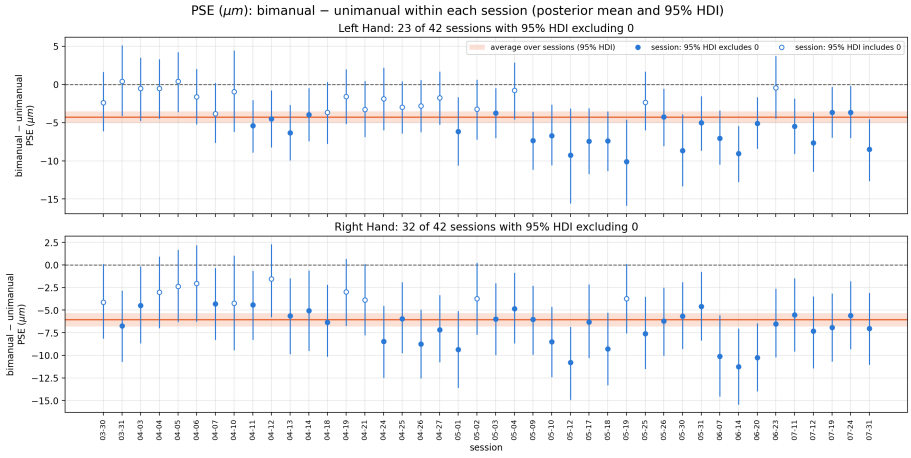
TODO: per-session PSE and JND reliable at ∼300 trials

Model B: fit to the real data



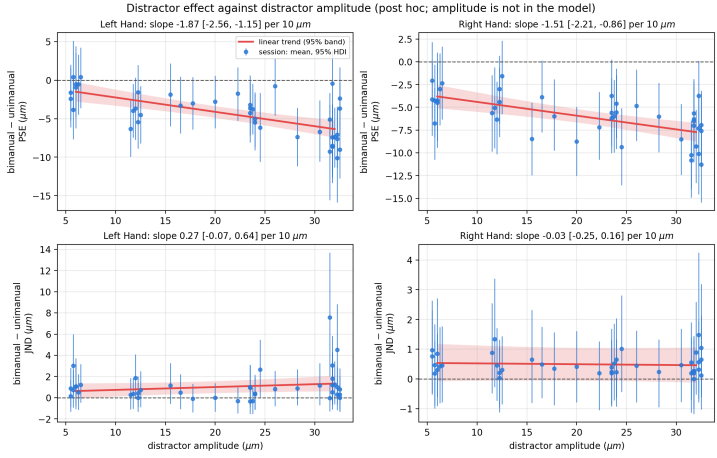
TODO: all 32 proportions inside; 99.8% of session cells (Fit_B_Report, "Posterior predictive checks")

Model B: PSE shift in every session



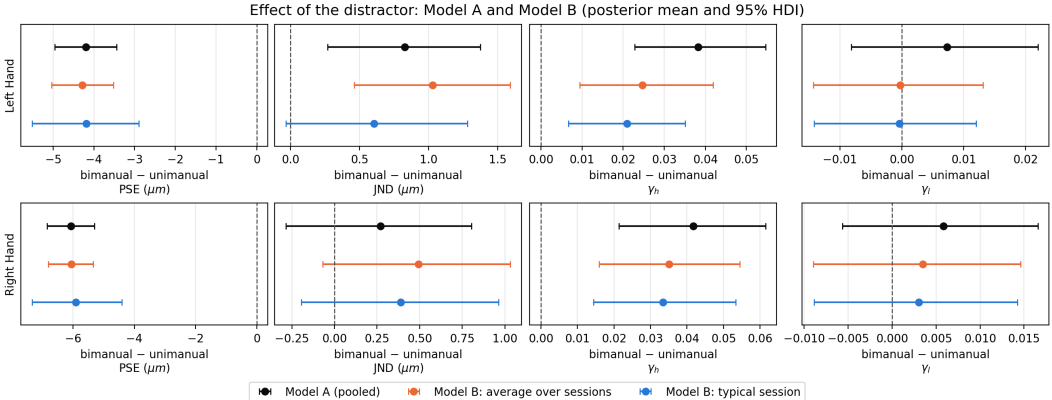
TODO: shift negative in 40/42 (left) and 42/42 (right) sessions (Fit_B_Report, Summary)

Model B: the shift grows with distractor amplitude (exploratory)



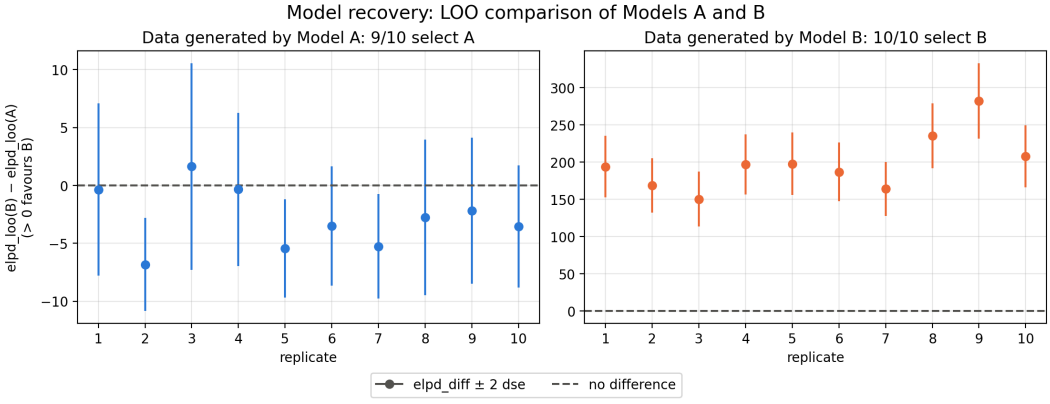
TODO: about -2 μm per 10 μm; confounded with session order

Model B against Model A



TODO: same effects; B better by LOO, 164 (SE 19) (Fit_B_Report, "Comparison with Model A")

Is B's advantage real? Model recovery



TODO: B wins on all data from B, ties or loses on data from A (Model_Recov_BA_Report)

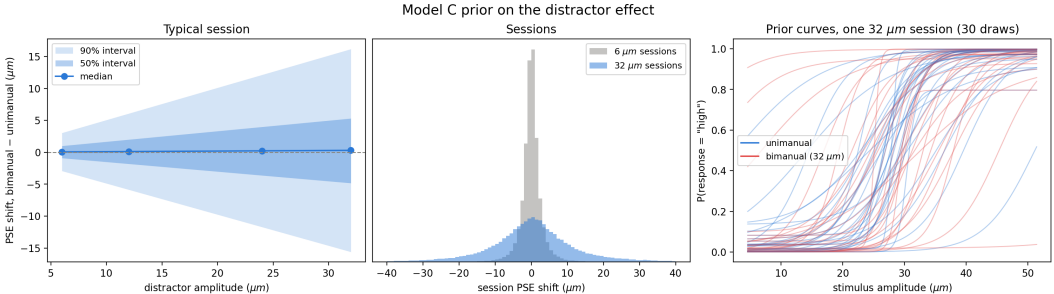
Model C

$$\psi(x, d) = \gamma_h + (1 - \gamma_h - \gamma_l)\ \text{logit}^{-1}(\beta_0 + \beta_1 x + \beta_2 d)$$

- d: distractor amplitude (0 on unimanual trials); one group per hand.
- Uni- and bimanual trials share slope and lapse rates: the distractor only shifts the curve, by an amount proportional to d.

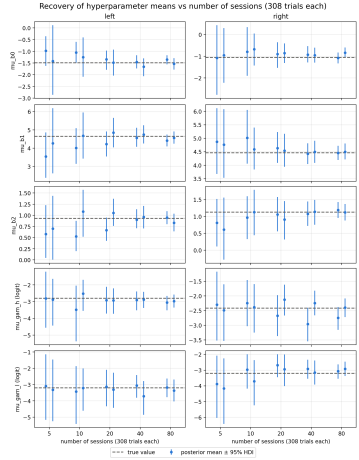
TODO: why: test the amplitude trend from Model B inside a model

Model C: prior on the distractor effect



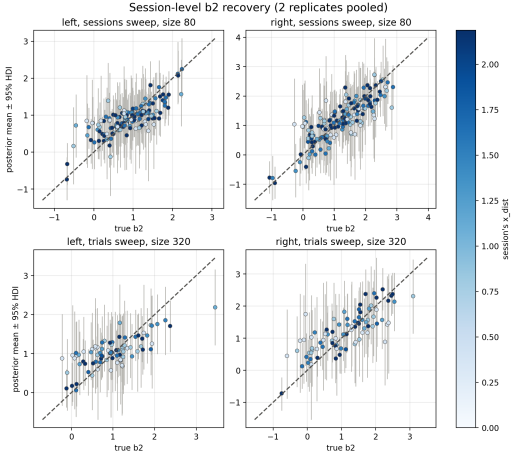
TODO: weakly informative; a weak distractor can only shift a little (Convergence_C_Report)

Model C: convergence on synthetic data



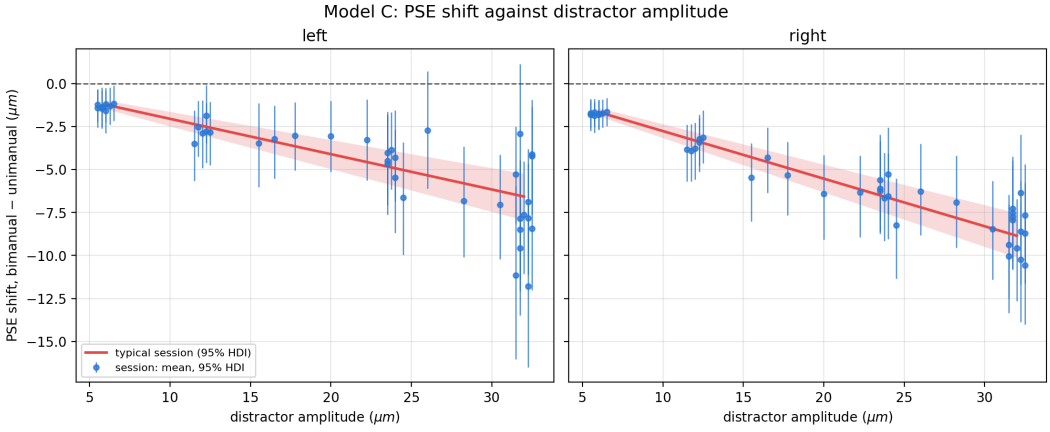
TODO: all hand-level means recovered, μ_{β_2} well (Convergence_C_Report, "Conclusions")

Model C: per-session β_2 is weak for small distractors



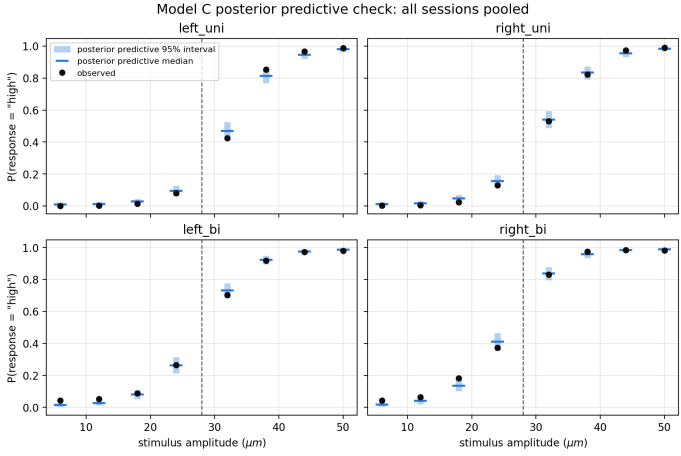
TODO: shrinkage toward the hand mean at 6 μm

Model C: the shift grows with distractor amplitude



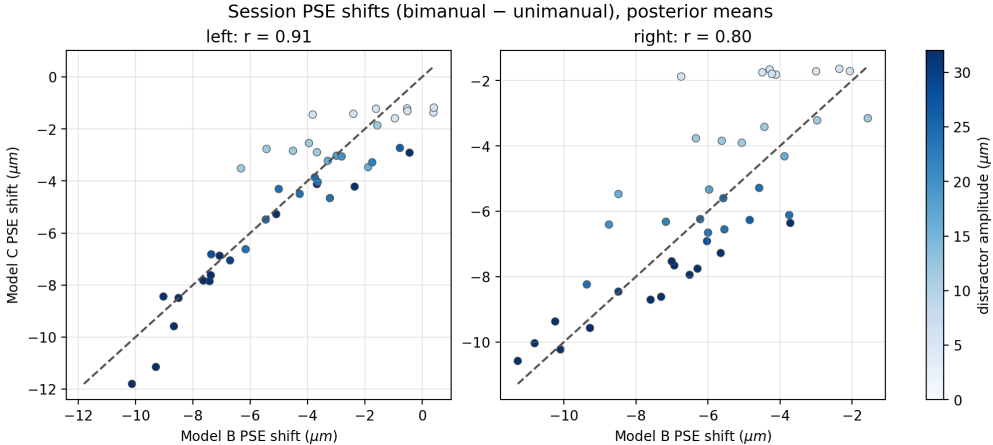
TODO: μ_{β_2} and the typical shift at 6 and 32 μm (Fit_C_Report, "Effect of the Distractor")

Model C misfits at the weakest stimuli



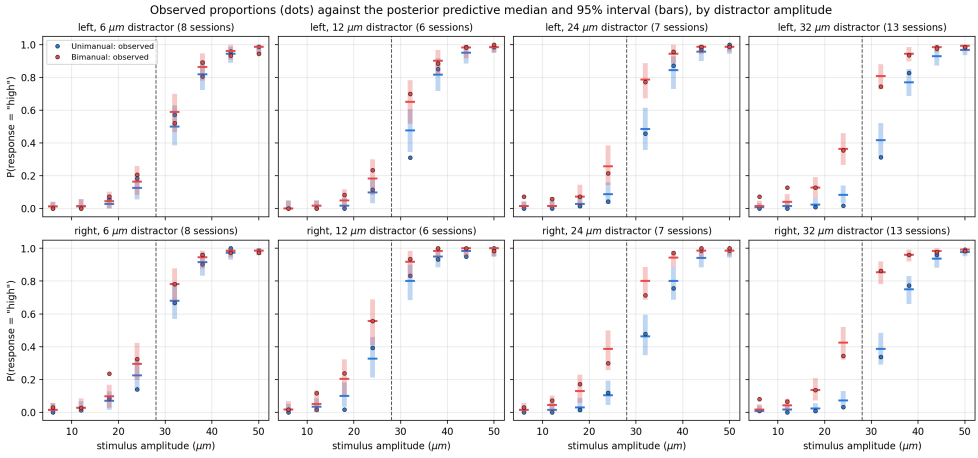
TODO: shared lapse rates: 6 of 32 proportions outside (Fit_C_Report, "Posterior predictive check")

Model C against Model B



TODO: LOO favors B by 37 (SE 12); 6 μm shifts too small (Fit_C_Report, Comparison)

Model C: shift by distractor amplitude, observed vs predicted



TODO: right hand at 6 μm outside: a presence effect (Fit_C_Report, amplitude check)

Comparison of the models

	A	B	C
Sessions	pooled	hierarchical	hierarchical
Distractor	separate groups	separate groups	covariate $\beta_2 d$
elpd_{loo}	**TODO:**	**TODO:**	**TODO:**
PSE shift, left / right (μm)	**TODO:**	**TODO:**	**TODO:**

TODO: fill from Fit_A/B/C_Report; elpd differences from Fit_B and Fit_C

Conclusions

- The distractor lowers the PSE: **TODO:** size, both hands.
- It raises the rate of "high" responses to weak stimuli (γ_h).
- The sessions differ; a hierarchical model is needed.
- The shift grows with distractor amplitude, **TODO:** but not proportionally?

Next steps

- Revise Model C: lapse rates per condition, a distractor-presence term, possibly a bimanual slope.
- Model recovery for B against the revised C.
- **TODO:** further data or experiments